NACKADEMIN · ETL COURSE

Loan Profit ETL

From a Riksbank rate to loan-level profit, on a schedule, with a paper trail.

A monthly SQL Server pipeline: ingest, validate, price, log, confirm.

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github.com/DavidFarm/loan-profit-etl

THE PROBLEM

An analyst ran the pricing queries by hand, every month.

A bank receives a vehicle file from an agency each month. Estimated profit per car depends on three things:

- **Energy class** of the car
- **Base rate**, which floats and changes monthly
- **Bank margin** added on top

```
profit = price × (base rate + margin) / 100
```

BEFORE

Manual queries, monthly

AFTER

```
SELECT * FROM
LoanProfitEstimates_202610;
```

Plus: logging, a confirmation mail with row counts, and a schedule.

ARCHITECTURE

One stored procedure, scheduled by SQL Agent.

Agency file
CarInformation_yyyymm

→

Staging
raw text, typed beside it

→

Validate
clean, or reject with a reason

→

LoanProfitEstimates_yyyymm
rebuilt in one transaction

Riksbank API
policy rate by month → InterestRates

EtlLog
every step, every warning

Database Mail
outcome, rows, table name

SQL Agent
1st of the month, 06:00

Dynamic table names for the month, built with `QUOTENAME` and parameterised `sp_executesql`.

SOURCE DATA

The brief gave no data. So I sourced it.

DATA	SOURCE	REAL?
Base interest rate	Riksbank open API, series SECBREPOEFF (policy rate), one value per month	Real
Models, years, fuel types	fueleconomy.gov vehicles.csv, mapped to Electric / Hybrid / Gasoline / Diesel	Real
Prices, customer risk tiers	Generated with a fixed random seed, so every run is reproducible	Synthetic
Bank margins	From the brief: 1.0 / 1.5 / 2.5 %	Given

Each monthly file deliberately contains messy rows: padded and lower-case classes, a missing price, a zero price, an unknown tier, an unknown energy class.

DESIGN DECISIONS

Built to fail in the right way.

DECISION	WHY
Text staging	A bad value in the file can never fail the load. It is cleaned or rejected with a reason.
Rate fallback, logged	No rate for the month: use the latest earlier one, and say which. No rate at all: hard stop.
Transactional rebuild	The analyst never sees half a table; a failed run leaves last month's result intact.
Mail outside the main try/catch	A broken mail server is a warning, not a failed run.
Failures re-thrown	SQL Agent history shows the job as failed when it is.
Re-runnable scripts	Every script and run is idempotent. Tested.

DATA QUALITY

180 rows in. 176 out. 4 rejected, and explained.

DBO.VW_RUNSUMMARY - LATEST RUNS

RunMonth	StartedAt	Status	RowsRead	RowsLoaded	RowsRejected	Warnings
2026-10-01	2026-10-25 19:26:13	SUCCESS WITH WARNINGS	180	176	4	3
2026-10-01	2026-10-25 19:24:02	SUCCESS WITH WARNINGS	180	176	4	3
2026-10-01	2026-10-25 19:09:49	SUCCESS WITH WARNINGS	180	176	4	4
2026-10-01	2026-10-25 19:04:33	SUCCESS WITH WARNINGS	180	176	4	4

DBO.VW_REJECTEDROWS - WHAT WAS REJECTED, AND WHY

RecordID	CarModel	EnergyClass	BasePrice	CustomerRiskTier	RejectReason
140	Cadillac XT4 AWD	Gasoline	0.00	D	Missing or non-positive BasePrice
167	Porsche Taycan GTS Sport Turismo	Steam	407470.82	C	Unknown EnergyClass
30	Volvo EX90 Twin Motor Performance (20 and 22 Inch Wheels)	Electric	781766.77	X	Unknown CustomerRiskTier
71	McLaren Automotive 765LT Spider	Gasoline		B	Missing or non-positive BasePrice

OUTPUT

What the analyst gets: one SELECT.

```
SELECT TOP 8 * FROM LOANPROFITESTIMATES_202610
```

RecordID	CarModel	EnergyClass	ManufactureYear	BasePrice	BaseInterestRate	BankMargin	TotalInterestRate	EstimatedProfit
1	Infiniti Q70 Hybrid	Hybrid	2015	335 687.25	1.75	1.50	3.25	10 909.84
2	Toyota Camry Hybrid LE	Hybrid	2021	500 921.30	1.75	1.50	3.25	16 279.94
3	Mercedes-Benz AMG EQS 4matic Plus	Electric	2024	586 506.12	1.75	1.00	2.75	16 128.92
4	Mazda CX-3 4WD	Gasoline	2017	219 711.11	1.75	2.50	4.25	9 337.72
5	Buick Enclave AWD	Gasoline	2018	401 472.42	1.75	2.50	4.25	17 062.58
6	Tesla Model S (85 kW-hr battery pack)	Electric	2015	388 127.54	1.75	1.00	2.75	10 673.51
7	Buick Encore AWD	Gasoline	2018	375 262.45	1.75	2.50	4.25	15 948.65
8	Porsche Taycan 4S Cross Turismo	Electric	2026	610 054.74	1.75	1.00	2.75	16 776.51

Every vehicle carries the base rate, the margin for its energy class, the total rate and the estimated profit.

CONFIRMATION

And what the team gets told.

CONFIRMATION EMAIL (LOCAL SMTP SINK)

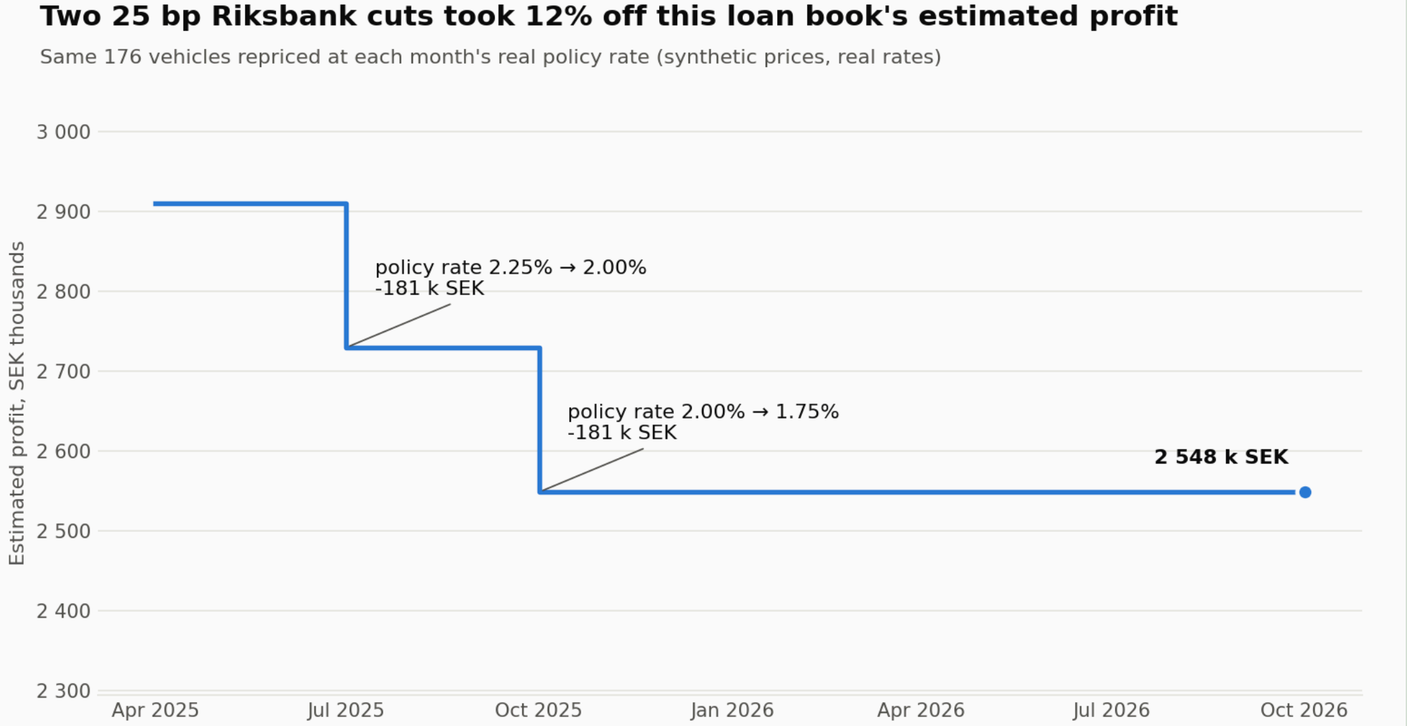
Subject: LoanETL 202610: SUCCESS with warnings
To: etl.analyst@loanetl.local

```
The LoanProfitEstimates job completed successfully.
Run month:        202610
Output table:     dbo.LoanProfitEstimates_202610
Rows processed:   180
Rows loaded:      176
Rows rejected:    4
Warnings:         3 (see dbo.EtlLog, RunID 6CC65480-6BB2-485C-995D-E270CF4CE1AD)
Query the result: SELECT * FROM LoanProfitEstimates_202610;
```

Captured by a local SMTP sink during development, so nothing left the machine.

RESULT

Two 25 bp cuts took 12.4% off estimated profit.



PORTFOLIO

176 vehicles

APR 2025 → OCT 2026

2 910k → 2 548k SEK

A view on top of the pipeline's own tables and formula. Real rates, synthetic prices.

TESTING AND OPERATIONS

8 end-to-end tests against a real SQL Server.

SCENARIO	WHAT IT PROVES
Rate present / missing	Normal run; exact fallback warning with month and value
Formula, normalisation	Profit matches the formula to the öre; " electric " becomes Electric
Rejects	Five bad rows, one per rule, rejected with the right reasons
Re-run, missing file, mail failure	No duplicates; loud failure with nothing created; warning, not failure

The SQL Agent job succeeded on its first manual start, and is scheduled for the 1st of each month.

SCOPE AND NEXT STEPS

What I'd build next.

- **Use the risk tier and depreciation tables.** Both are in the data model; the specified formula doesn't use them yet.
- **Alert on failure.** Route Agent job failures to an operator, not only to the log.
- **Real SMTP and a service account.** The pipeline mails a local sink today.
- **Loan terms.** Price by term and amortisation instead of a flat rate.

Started as an ETL assignment on the Data Engineer programme at Nackademin. The data layer, quality views, tests and CI are my additions.